INCH-POUND

MIL-R-52807B
22 August 1991
SUPERSEDING
MIL-R-52807A(ME)
16 August 1982

MILITARY SPECIFICATION

ROLLER, TRANSOM, BAILEY BRIDGE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of transom roller for use in Bailey bridge erection.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

QQ-C-390	-	Copper Alloy Castings (Including Cast Bar).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MS-24665	-	Pin, Cotter (Split).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel.
A 53 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated Welded and Seamless.
A 108 - Steel Bars, Carbon, Cold-Finished, Standard Quality.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The roller shall be as shown in figure 1 and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3).

3.3 Material. Material shall be as specified herein.

3.3.1 Material deterioration prevention and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the Bailey bridge transom roller may be newly fabricated from recovered materials to the maximum extent practicable, provided the item produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the item.

3.3.3 Structural steel. Structural steel shall conform to ASTM A 36, 12 pounds.

3.3.4 Cotter pins. Cotter pins shall conform to MS-24665-497. Each roller shall be provided with an extra cotter pin.

3.3.5 Steel bar. Steel for pins and bars shall conform to ASTM A 108, UNS G10200, cold drawn and stress relieved.

3.3.6 Pipe. Pipe shall conform to ASTM A 53, standard weight, except that ends shall not be threaded and hydrostatic test is not required.

3.3.7 Bronze. Bronze for bushings shall conform to QQ-C-390, copper alloy no. C90500, type I, condition as cast.

3.4 Identification marking. The roller shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The portions of the roller normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type A. Color shall be in accordance with FED-STD-595, color no. 34083.

3.6 Workmanship.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.6.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the transom rollers to conform to the examination requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the

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welding process used in fabrication of the transom rollers shall be at the option of the contractor. Welding procedures shall be qualified in accordance with applicable ASME or AWS code, and all welding and welds shall be in accordance with ASME or AWS code.

3.6.3 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX.
AWS D1.1 - Structural Welding Code - Steel.

Contractors who make only horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The roller shall be examined as specified herein for the following defects:

Major

101. Roller not in accordance with figure 1 (see page 9).
102. Materials not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
104. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified (see 3.3.1.1).
105. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.1.2).
106. Workmanship not as specified (see 3.6).
107. Welding not as specified (see 3.6.2).
108. Welders and welding operators not certified as specified (see 3.6.3).

Minor

201. Identification marking missing, incomplete, or illegible (see 3.4).
202. Treatment and painting not as specified (see 3.5).

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. One or more defects shall be cause for rejection.

109.	Shipping containers not as specified for levels A, B or C	(see 5.1.1, 5.1.2 and 5.1.3).
110.	Quantity packed together exceeds the weight or size limitation of the box for level A, B or C	(see 5.1.1, 5.1.2 and 5.1.3).
111.	Rollers not blocked and braced in accordance with the referenced document as specified for level A or B	(see 5.1.1 and 5.1.2).
112.	Marking missing, illegible, incorrect, or incomplete for level A, B or C	(see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The rollers shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. The contents shall be blocked and braced within the box to prevent free movement or damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to ASTM D 3953 type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.1.2 Level B. The rollers shall be packed as specified in 5.1.1, except boxes shall be domestic type. As an alternate, a box conforming to PPP-B-636, V3c, V11c, V13, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitations of the box. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be as specified in 5.1.1.

5.1.3 Level C. The roller shall be packed as specified for level B, in boxes conforming to PPP-B-636 except the boxes shall be domestic class and strapping shall not be zinc-coated.

5.2 Marking (see 6.4). In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The transom roller is intended for use in the erection of portable panel Bailey-type bridges. This is a component part of the bridge set listed in Supply Catalog (SC); SC 5420-97-CL-E40

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and numbers of units required (see 3.2).
d. Welding process required when other than as specified (see 3.6.2).
e. Level of packing required (see 5.1).
f. Any special marking required (see 5.2).

6.3 First article. When a first article inspection is required, the transom roller should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Preservation/packing level marking. Preservation is not required. For purposes of preservation/packing level marking, preservation shall be designated level A.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term key word listings.

Bridge, Bailey
Roller
Transom

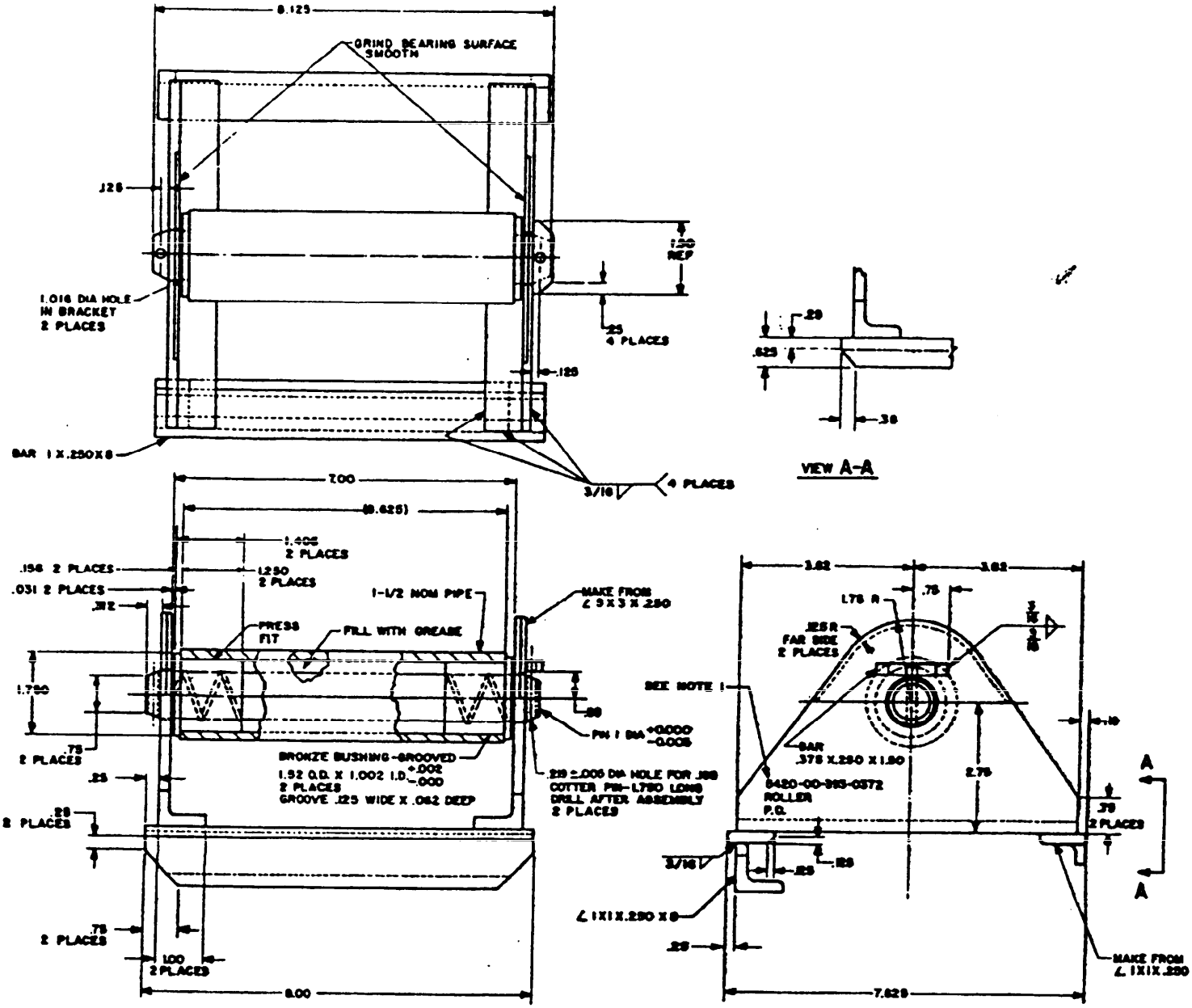
Custodian:
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Preparing activity:
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Review Activity:
DLA-CS

Project 5420-0188

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NOTES:

1. STEEL STAMP, .250 HIGH CHARACTERS, AT APPROXIMATE LOCATION SHOWN.
2. ALL DIMENSIONS ARE IN INCHES.
3. TOLERANCES, UNLESS OTHERWISE SHOWN: .XX ±.06 .XXX±.030

FIGURE 1. Transom roller.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given.
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RECOMMENDED CHANGE:	1. DOCUMENT NUMBER MIL-R-52807B(ME)	2. DOCUMENT DATE (YYMMDD) 910822

3. DOCUMENT TITLE
Roller, Transom, Bailey Bridge

4. NATURE OF CHANGE (identify paragraph number and include proposed if possible. Attach extra sheets as needed).

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (LAST,FIRST MIDDLE INITIAL)	b. ORGANIZATION	
c. ADDRESS (Include zip code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Name Gerry Harlow	b. TELEPHONE (In lude Area Code) (1) Commercial (2) AUTOVON (703) 664-2071 354-2071
c. ADDRESS (Include zip code) Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	If you do not receive a reply with 45 days, contact: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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